

Notes: Miller and Weinberg (Econometrica, 2017)

Understanding the Price Effects of the MillerCoors Joint Venture

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1 Big picture

- **Question.** Can the large beer price increases observed after the MillerCoors joint venture be explained by standard **unilateral merger effects** under Nash–Bertrand competition, or did the merger also facilitate **post-merger coordination**?
- **Setting.** The paper studies the 2008 **MillerCoors joint venture** in the U.S. brewing industry using retail beer prices across **39 geographic regions** from **2001–2011**, together with a structural differentiated-products demand-and-supply model.
- **Core challenge.** In standard merger simulation, firms are assumed to play one-shot Nash–Bertrand both before and after the merger. But here, not only MillerCoors prices rose after the merger — **Anheuser-Busch InBev (ABI)** prices rose too. That is hard to reconcile with unilateral effects alone.
- **Main methodological contribution.** The paper augments a standard merger model with a **post-merger conduct parameter** κ that captures the extent to which **ABI and MillerCoors internalize each other’s pricing externalities** after the merger. This lets the authors test whether post-merger pricing still looks Nash–Bertrand.
- **Why it matters.** Merger policy often focuses on **unilateral effects**, but antitrust agencies are also concerned that mergers in concentrated markets can make **coordination easier**. This paper provides a structural way to test that idea ex post.
- **Main empirical question.** Did the MillerCoors joint venture soften competition only through unilateral internalization between Miller and Coors, or also through coordination between MillerCoors and ABI?
- **Main empirical results.**
 - Retail prices of MillerCoors and ABI rise abruptly after the merger, while imported rivals such as **Corona** and **Heineken** do not show the same persistent increase.
 - A descriptive difference-in-differences exercise implies a price increase of roughly **six percent** after the merger.
 - The supply-side conduct parameter is **positive and significant**; with the preferred demand specification, it implies that ABI and MillerCoors internalize about **26%** of their pricing externalities after the merger.
 - Counterfactual simulations imply that observed post-merger prices are about **6%–8% higher** than under post-merger Nash–Bertrand competition.
 - Markups are about **17%–18% higher** than under post-merger Nash–Bertrand competition.
 - The merger reduces **consumer surplus by 3.7%** relative to the no-merger benchmark, but with efficiencies included it still increases **total surplus by 1.3%**.
- **Economic takeaway.** In this market, **merger efficiencies and unilateral effects roughly offset each other**, while **coordinated effects** explain most of the observed post-merger price increase and most of the consumer harm.

2 Data and measurement (key objects)

2.1 Industry background and market structure

- **Industry.** The paper studies U.S. brewing, a branded consumer-products industry with state alcohol regulations that create a three-tier structure: brewers sell to **state-licensed distributors**, who then sell to retailers.
- **Vertical institutions.**
 - Large brewers cannot directly sell to retail outlets.
 - Payments such as slotting fees and slotting allowances are prohibited in this supply chain.
 - Retail price maintenance is formally illegal in many states, but the paper notes that distributors are often induced to sell at brewer-set wholesale prices.
- **Main firms.** Five brewers dominate the national market in the sample period:
 - **ABI**,
 - **SABMiller**,
 - **Molson Coors**,
 - **Grupo Modelo**,
 - **Heineken**.
- **Concentration.**
 - These five firms account for roughly **80%** of total retail revenue throughout the sample.
 - ABI accounts for about **35%** of retail revenue.
 - MillerCoors accounts for about **28%–30%** after the joint venture.
 - Modelo and Heineken together account for about **15%** of revenues.
 - National HHIs are in the **moderately concentrated** range, but local concentration is often much higher: **23 of the 39 regions** are above 2,500 in 2011.

2.2 Data sources

- **Main retail data.** The primary source is the **IRI Academic Database**, which contains weekly store-level scanner data on revenue and unit sales for supermarkets over **2001–2011**.
- **Sample restrictions.** The regression samples use:
 - **39 geographic regions**,
 - **13 flagship brands**: Bud Light, Budweiser, Michelob, Michelob Light, Miller Lite, Miller Genuine Draft, Miller High Life, Coors Light, Coors, Corona Extra, Corona Extra Light, Heineken, and Heineken Light.
- **Excluded brands.** The omitted brands are mainly regional or low-price subpremium products.

- **Consumer demographics.** The paper uses the **American Community Survey PUMS** for **2005–2011**, drawing 500 households per region-year. Household income is total household income divided by household size. Mean income is about **\$38,000**.
- **Transportation-cost data.**
 - Driving miles from each IRI region to the nearest brewery are obtained with **Google Maps**.
 - For imports, the relevant distance is from the region to the nearest relevant port (or in Modelo’s case, to Ciudad Obregon in Mexico).
 - Distance is interacted with the **diesel fuel price index** to proxy for marginal shipping costs.

2.3 Markets, products, and key constructed objects

- **Product definition.** A product is a **brand** $\times$ **package size** combination.
- **Package sizes.** The paper focuses on:
 - 6 packs,
 - 12 packs,
 - 24 packs,
 - 30 packs.

The 24-pack and 30-pack formats are combined because their use varies by region.

- **Unit of quantity.** Market shares are computed in **144-ounce equivalent units**, corresponding to a 12-pack. So a 6-pack is weighted by 50%.
- **Price measure.** Prices are defined as **revenue divided by equivalent unit sales**. Larger package sizes are generally cheaper on an equivalent-unit basis.
- **Aggregation.** For estimation, the weekly store-level data are aggregated to:
 - **region–month**, and
 - **region–quarter**
 panels.
- **Observation counts.**
 - The broader descriptive sample contains **167,695** product–region–month–year observations.
 - The structural demand and supply samples contain **94,656** monthly observations and **31,784** quarterly observations.
 - The months from **June 2008 through May 2009** are excluded in the structural analysis so that merger-related cost reductions have time to materialize.
- **Market size.** The outside good market size is assumed to be **50% greater than the maximum observed unit sales** in each region.

2.4 Key descriptive patterns

- **Raw prices.** Before the merger, prices for domestic and imported beers trend downward. After the merger, prices of **Bud Light**, **Miller Lite**, and **Coors Light** jump upward by about 8%, while **Corona Extra** and **Heineken** keep trending downward.
- **Difference-in-differences.**
 - In the flagship 12-pack sample, MillerCoors and ABI prices rise by about 6.8% and 5.7% in absolute terms.
 - In broader samples the price increases are around 5% for MillerCoors and 4% for ABI.
 - When the first post-merger year is excluded, these estimates rise further, to around 6%–7%.
- **Cross-sectional variation.**
 - Regions with larger predicted concentration increases (higher ΔHHI) have larger price increases.
 - Regions with larger merger-related shipping distance reductions have smaller price increases.
 - These two forces nearly offset on average, leaving an unexplained common post-merger increase of roughly 4%–5% for MillerCoors and ABI.

Interpretation. The raw and reduced-form evidence lines up with unilateral effects in the cross section, but the large common post-merger increase — especially for ABI — suggests that unilateral effects are not the whole story.

3 Models and empirical specifications (all equations + interpretation)

3.1 Consumer demand: random coefficient nested logit

Consumer i 's conditional indirect utility from product j in region r and period t is

$$u_{ijrt} = x_j \beta_i^* + \alpha_i^* p_{jrt} + \sigma_j^D + \tau_t^D + \xi_{jrt} + \varepsilon_{ijrt},$$

where

$$\begin{bmatrix} \alpha_i^* \\ \beta_i^* \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} + \Pi D_i.$$

- x_j includes a constant, calories, package size, and an import indicator.
- p_{jrt} is the retail price.
- σ_j^D is a product fixed effect.
- τ_t^D is a time fixed effect.
- ξ_{jrt} is an unobserved product–region–time demand shock.
- D_i is consumer income.

The nested-logit error structure is

$$\varepsilon_{ijrt} = \zeta_{igrt} + (1 - \rho)\epsilon_{ijrt},$$

with $0 \leq \rho < 1$.

Interpretation. The model lets products within the inside-good nest be closer substitutes than the outside good. The random-coefficient term ΠD_i allows consumer income to shift price sensitivity and tastes for observed product characteristics, which is how the model breaks the restrictive proportional-substitution pattern of simple logit.

3.2 Mean utility, market shares, and the nested-logit special case

The paper decomposes indirect utility into mean utility and consumer-specific deviations:

$$\delta_{jrt} = x_j\beta + \alpha p_{jrt} + \sigma_j^D + \tau_t^D + \xi_{jrt},$$

$$\mu_{ijrt} = [p_{jrt}, x_j]^\top \Pi D_i.$$

The market share equation is

$$s_{jrt} = \frac{1}{N_{rt}} \sum_{i=1}^{N_{rt}} \frac{\exp((\delta_{jrt} + \mu_{ijrt})/(1 - \rho)) \exp(I_{igrt})}{\exp(I_{igrt}/(1 - \rho)) \exp(I_{irt})}.$$

If $\Pi = 0$, the model collapses to a nested logit:

$$\log(s_{jrt}) - \log(s_{0rt}) = x_j\beta + \alpha p_{jrt} + \sigma_j^D + \tau_t^D + \rho \log(s_{jrt|g}) + \xi_{jrt}.$$

Interpretation. The nested-logit benchmark already makes substitution within beer stronger than substitution to the outside good. The full RCNL specification goes further by making substitution patterns depend on income and product characteristics, which is important for getting credible unilateral-effects predictions.

3.3 Demand estimation and instruments

Demand is estimated with a **nested fixed-point GMM procedure** in the spirit of Berry, Levinsohn, and Pakes. The population moment condition is

$$E[Z \cdot \omega(\theta_0^D)] = 0,$$

and the estimator is

$$\hat{\theta}^D = \arg \min_{\theta} \omega(\theta)^\top Z A^{-1} Z^\top \omega(\theta).$$

The main instruments are:

- **Price instruments:**
 - distance from brewery to region interacted with diesel prices,
 - an indicator for ABI and MillerCoors products after the merger.

- **Nested-logit instruments:**
 - number of products in the market,
 - total distance summed across competing products,
 - interactions of those variables with ABI and MillerCoors indicators.
- **Heterogeneity instruments:** mean income interacted with observed product characteristics.

Interpretation. Distance is a cost shifter, the merger indicator is an excluded shift in market structure, and the product-count / total-distance instruments create exogenous variation in the conditional share term that identifies the nesting parameter. Income interactions identify how tastes vary across households.

3.4 Demand estimates and substitution patterns

The preferred RCNL specifications imply:

- median own-price elasticities roughly between -4.33 and -6.10 ,
- market-price elasticities much smaller in magnitude,
- outside diversion of roughly **12%–17%** in the RCNL models.

Interpretation. Most substitution occurs **within beer products**, not to the outside good. Consumers of a product such as Bud Light disproportionately substitute toward similar domestic lagers and nearby package sizes, while consumers of imports such as Corona disproportionately substitute toward Heineken and smaller packs. These substitution patterns are what determine the unilateral price effects of the merger.

3.5 Supply-side pricing model

Brewers choose prices according to the differentiated-products first-order condition

$$p_t = mc_t - \left[\Omega_t(\kappa) \circ \left(\frac{\partial s_t(p_t; \theta^D)}{\partial p_t} \right)^\top \right]^{-1} s_t(p_t; \theta^D),$$

where:

- p_t is the vector of prices,
- mc_t is the vector of marginal costs,
- $s_t(\cdot)$ is the vector of market shares,
- $\Omega_t(\kappa)$ is the ownership/conduct matrix,
- $\circ$ denotes element-by-element multiplication.

In the post-merger periods, κ captures the extent to which **ABI** and **MillerCoors** internalize each other's pricing externalities. The model nests:

- **post-merger Nash–Bertrand** if $\kappa = 0$,

- **joint profit maximization between ABI and MillerCoors** if $\kappa = 1$.

Interpretation. This is the central conduct test. The merger mechanically changes ownership between Miller and Coors, generating standard unilateral effects. The extra parameter κ asks whether *ABI* also behaves as if it partially internalizes the merged firm’s pricing incentives.

3.6 Ownership matrix and economic meaning of κ

In a simple example with products sold by ABI, Miller, Coors, and Modelo, the pre- and post-merger ownership matrices are

$$\Omega_{t_1^*} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \Omega_{t_2^*} = \begin{bmatrix} 1 & \kappa & \kappa & 0 \\ \kappa & 1 & 1 & 0 \\ \kappa & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Interpretation.

- Miller and Coors fully internalize one another after the merger.
- κ governs the extra post-merger interaction between the merged MillerCoors entity and ABI.
- Modelo and Heineken are restricted to compete *à la* Nash throughout.

3.7 Marginal costs

The marginal-cost specification is

$$\text{mc}_{jrt} = w_{jrt}\gamma + \sigma_j^S + \tau_t^S + \mu_r^S + \eta_{jrt},$$

where w_{jrt} includes:

- **distance** (miles $\times$ diesel index), and
- a **MillerCoors $\times$ post-merger** indicator.

Interpretation. The distance term captures shipping-related marginal costs. The MillerCoors post-merger indicator captures merger-specific cost synergies that are not fully explained by shorter shipping distances — for example, distribution rationalization and other marginal-cost savings.

3.8 Supply estimation and identification

The supply side is estimated by method of moments using

$$E[z^\top \eta^*(\theta_0^S)] = 0,$$

with estimator

$$\hat{\theta}^S = \arg \min_{\theta} \eta^*(\theta; \hat{\theta}^D)^\top z z^\top \eta^*(\theta; \hat{\theta}^D).$$

The excluded supply-side instrument is an indicator equal to one for **ABI and MillerCoors in the post-merger periods**.

Interpretation. Identification comes from asking whether post-merger ABI prices are *too high* relative to what unilateral effects and common demand/cost shocks would predict. If they are, the model attributes the excess to a positive κ .

3.9 Counterfactual design

Using the preferred RCNL-1 demand estimates, the paper recomputes equilibrium under five scenarios:

1. observed post-merger world: unilateral effects + coordinated effects + efficiencies;
2. coordinated effects + unilateral effects, but **no efficiencies**;
3. unilateral effects + efficiencies, but **no coordinated effects**;
4. unilateral effects only (no efficiencies, no coordination);
5. **no merger**.

Interpretation. These counterfactuals separate three distinct forces:

- **unilateral effects** from internalization between Miller and Coors,
- **coordinated effects** captured by $\kappa > 0$,
- **cost efficiencies** captured through distance savings and common marginal-cost shifts.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard merger approach

- Standard ex post merger simulation keeps the equilibrium concept fixed at **Nash–Bertrand**.
- Here, that model cannot explain the fact that **ABI** — the main rival, not a merger participant — also raises prices sharply after the merger.
- If the authors force post-merger Nash–Bertrand on the data, ABI would need an implausibly large **cost increase of about 13%** to rationalize observed prices.

Interpretation. The key failure is not just that prices rise, but that the rival’s prices rise too much relative to what unilateral diversion alone should generate.

4.2 What identifies the new approach

- **Demand side:** the RCNL model identifies realistic elasticities and diversion ratios, which pin down the magnitude of unilateral effects.
- **Cost side:** distance and the MillerCoors post-merger cost shifter identify merger efficiencies.
- **Conduct side:** the post-merger ABI/MillerCoors indicator instruments for the markup term and identifies whether ABI’s prices move beyond unilateral predictions.

Key identifying assumption. Changes in ABI’s unobserved demand and costs before versus after the merger are not systematically different from the corresponding unobserved changes for **Modelo and Heineken**. Those firms serve as the outside comparison group.

4.3 Why the ABI comparison is so useful

- MillerCoors experiences both unilateral incentives and merger efficiencies, so its price response alone is hard to interpret.
- ABI does *not* receive merger-specific efficiencies from the Miller/Coors transaction.
- Therefore, when ABI prices rise too much, the cleanest interpretation is that competition has softened more broadly.

Interpretation. ABI is effectively the “canary in the coal mine” for coordination: if ABI’s price increase is too large, then the post-merger equilibrium is not just standard unilateral pricing.

4.4 Empirical credibility of the application

- The reduced-form cross-sectional evidence lines up with the supply-side forces: larger local Δ HHI increases push prices up, while shorter shipping distances push them down.
- The qualitative evidence matches the model’s structure: ABI and MillerCoors appear to follow each other’s annual price announcements, while Modelo appears to pursue a strategy of narrowing price gaps rather than joining the coordinated increases.
- The results are robust across nested-logit and RCNL demand specifications and across monthly versus quarterly data.
- Even when pre-merger conduct is normalized away from Nash, the estimated post-merger conduct parameter still rises significantly, so the key result is a **change in conduct**, not just a level assumption.

4.5 Caveats and limitations

- The merger coincides with the **Great Recession**, so other macro changes may have contributed to pricing shifts.
- If the model understates strategic complementarity under Nash pricing — for example because firms signal costs dynamically — then some coordinated effects could be overstated.
- The paper does not offer a fully specified theory of *why* the merger facilitated coordination; instead, it documents that the observed price changes are most consistent with that interpretation.

4.6 Relation to the literature

- Connects to **merger retrospectives** that compare realized post-merger outcomes with standard merger simulation.

- Connects to the **conduct-parameter** literature that uses regime shifts to infer departures from standard competition.
- Extends the empirical merger literature by allowing the equilibrium concept itself to change after a merger.

5 Tables and figures

5.1 Table I and Figure 1: market structure and raw price paths

TABLE I
REVENUE SHARES AND HHI^a

Year	ABI	MillerCoors	Miller	Coors	Modelo	Heineken	Total	HHI
2001	0.37	—	0.20	0.12	0.08	0.04	0.81	2,043
2003	0.39	—	0.19	0.11	0.08	0.05	0.82	2,092
2005	0.36	—	0.19	0.11	0.09	0.05	0.79	1,907
2007	0.35	—	0.18	0.11	0.10	0.06	0.80	1,853
2009	0.37	0.29	—	—	0.09	0.05	0.80	2,350
2011	0.35	0.28	—	—	0.09	0.07	0.79	2,162

^aThis table provides revenue shares and the HHI over 2001–2011. Firm-specific revenue shares are provided for ABI, Miller, Coors, Modelo, and Heineken. The total across these firms is also provided. The HHI is scaled from 0 to 10,000. The revenue shares incorporate changes in brand ownership during the sample period, including the merger of Anheuser-Busch (AB) and InBev to form ABI, which closed in November 2008, and Heineken's acquisition of the FEMSA brands in April 2010. All statistics are based on supermarket sales recorded in IRI scanner data.

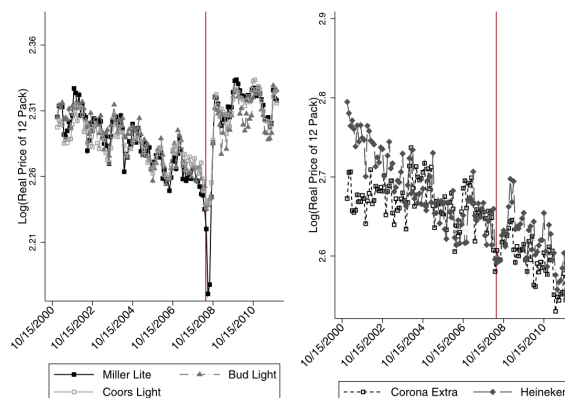


FIGURE 1.—Average retail prices of flagship brand 12 packs. *Notes:* This figure plots the average prices of a 12 pack over 2001–2011, separately for Bud Light, Miller Lite, Coors Light, Corona Extra, and Heineken. The vertical axis is the natural log of the price in real 2010 dollars. The vertical bar drawn at June 2008 signifies the consummation of the Miller–Coors merger.

Read:

- Table I shows that the leading five brewers account for about **80%** of revenue, with ABI around **35%** and MillerCoors around **30%**. National concentration is moderate, but many local markets are highly concentrated.
- Figure 1 is the cleanest reduced-form picture in the paper. After the June 2008 merger, **Bud Light, Miller Lite, and Coors Light** move up sharply, while **Corona Extra** and **Heineken** do not show the same persistent increase.
- The most important visual point is that **a rival's price rises together with the merged firm's prices**. That is what makes coordination plausible.

5.2 Table II: difference-in-differences price increases

Read:

- In the narrow 12-pack flagship sample, MillerCoors and ABI prices increase by about **10.3%** and **9.1%** relative to imports, corresponding to absolute increases of roughly **6.8%** and **5.7%**.
- In the broader sample, the estimated absolute post-merger price increases are about **5%** for MillerCoors and **4%** for ABI.
- Excluding the first post-merger year makes these effects even larger.

- This table establishes the basic fact to be explained by the structural model.

TABLE II
CHANGES IN RETAIL PRICES BY FIRM^a

	(i)	(ii)	(iii)	(iv)
1{MillerCoors} × 1{ Post-Merger}	0.098 (0.007)	0.050 (0.004)	0.047 (0.005)	0.069 (0.007)
1{ABI} × 1{ Post-Merger}	0.087 (0.007)	0.040 (0.005)	0.038 (0.005)	0.062 (0.007)
1{Post-Merger}	-0.031 (0.005)	-0.007 (0.004)	-0.002 (0.004)	0.010 (0.009)
log(Employment)	-	-	-0.051 (0.080)	0.131 (0.081)
log(Earnings)	-	-	0.156 (0.029)	0.152 (0.035)
Pre-Merger Average Price	11.75	11.14	11.14	11.14
Product Trends	No	No	Yes	Yes
Covariates	No	No	Yes	Yes
# Observations	25,740	167,695	167,695	151,525

^aEstimation is with OLS. The dependent variable is log real retail price. Observations are at the brand-size-region-month-year level. Column (i) includes 12 packs of Bud Light, Coors Light, Miller Lite, Corona Extra, and Heineken. Columns (ii) and (iii) includes the six-, 12, and 24 packs of these brands plus Budweiser, Michelob Ultra, Coors, Miller Genuine Draft, Miller High Life, Corona Light, and Heineken Premium Light. The estimation sample spans 39 regions from 2001 to 2011, except in column (iv), which excludes June 2008 through May 2009. All regressions include a linear time trend and product (brand × size) fixed effects interacted with region fixed effects. Standard errors are clustered at the region level and shown in parentheses.

TABLE III
CROSS-SECTIONAL VARIATION IN PRICE INCREASES^a

	Pooled	MillerCoors	ABI	Imports
$\Delta HHI \times 1\{\text{Post-Merger}\}$	0.997 (0.454)	1.172 (0.542)	1.503 (0.531)	-0.005 (0.534)
$\Delta \text{MILES} \times 1\{\text{Post-Merger}\}$	-0.042 (0.013)	-0.040 (0.016)	-0.053 (0.013)	-0.028 (0.014)
1{Post-Merger}	0.037 (0.012)	0.049 (0.014)	0.040 (0.013)	0.019 (0.014)
# Observations	167,695	75,315	50,810	41,570

^aEstimation is with OLS. The dependent variable is log real retail price. Observations are at the brand-size-region-month-year level. The estimation sample spans 39 regions from 2001 to 2011. All regressions include a linear trend and product (brand × size) fixed effects interacted with region fixed effects. Standard errors are clustered at the region level and shown in parentheses.

5.3 Table III and the documentary record: unilateral patterns plus a common post-merger shift

Read:

- Table III shows that price increases are larger in regions with bigger ΔHHI changes and smaller where shipping distances fall more. That is exactly what unilateral effects plus efficiencies would predict.
- But the average effects of these two forces nearly cancel out. A remaining common post-merger shift of about 4%–5% is left over for MillerCoors and ABI.
- The paper complements this with qualitative evidence: ABI’s pricing documents emphasize followership and transparent pricing plans, while Modelo appears not to follow the coordinated increases.

5.4 Table IV and Table V: demand estimates and substitution patterns

TABLE IV
BASELINE DEMAND ESTIMATES^a

Demand Model: Data Frequency: Variable	Parameter	NL-1 Monthly (i)	RCNL-1 Monthly (ii)	RCNL-2 Quarterly (iii)	RCNL-3 Monthly (iv)	RCNL-4 Quarterly (v)
Price	α	-0.1312 (0.0884)	-0.0887 (0.0141)	-0.1087 (0.0163)	-0.0798 (0.0147)	-0.0944 (0.0146)
Nesting Parameter	ρ	0.6299 (0.0941)	0.8299 (0.0402)	0.7779 (0.0479)	0.8079 (0.0602)	0.8344 (0.0519)
<i>Demographic Interactions</i>						
Income × Price	Π_1		0.0007 (0.0002)	0.0009 (0.0003)		
Income × Constant	Π_2		0.0143 (0.0051)	0.0125 (0.0055)	0.0228 (0.0042)	0.0241 (0.0042)
Income × Calories	Π_3		0.0045 (0.0016)	0.0045 (0.0017)	0.0038 (0.0018)	0.0031 (0.0015)
Income × Import	Π_4				0.0039 (0.0019)	0.0031 (0.0016)
Income × Package Size	Π_5				-0.0013 (0.0007)	-0.0017 (0.006)
<i>Other Statistics</i>						
Median Own Price Elasticity		-3.81	-4.74	-4.33	-4.45	-6.10
Median Market Price Elasticity		-1.10	-0.60	-0.72	-0.60	-0.69
Median Outside Diversion		29.80%	12.96%	16.98%	13.91%	11.82%
J-Statistic			13.94	13.75	13.91	14.15

^aThis table shows the baseline demand results. We use 2SLS for estimation in column (i) and GMM in columns (ii) to (v). There are 94,656 observations at the brand-size-region-month-year level in columns (i), (ii), and (iv) and 31,784 observations at the brand-size-region-year-quarter level in columns (iii) and (v). The samples exclude the months/quarters between June 2008 and May 2009. All regressions include product (brand × size) and period (month or quarter) fixed effects. The elasticity and diversion numbers represent medians among all the brand-size-region-month/quarter-year observations. Standard errors are clustered by region and shown in parentheses.

TABLE V
MEAN ELASTICITIES FOR 12-PACK PRODUCTS FROM SPECIFICATION RCNL-1^a

Brand/Category	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
<i>Product-Specific Own and Cross-Elasticities</i>													
(1) Bud Light	-4.389	0.160	0.019	0.182	0.235	0.101	0.146	0.047	0.040	0.130	0.046	0.072	0.196
(2) Budweiser	0.323	-4.272	0.019	0.166	0.258	0.103	0.166	0.047	0.039	0.121	0.043	0.068	0.183
(3) Coors	0.316	0.154	-4.371	0.163	0.259	0.102	0.167	0.046	0.038	0.119	0.042	0.066	0.180
(4) Coors Light	0.351	0.160	0.019	-4.628	0.230	0.100	0.142	0.047	0.041	0.132	0.047	0.073	0.199
(5) Corona Extra	0.279	0.147	0.018	0.137	-5.178	0.108	0.203	0.047	0.035	0.104	0.035	0.061	0.158
(6) Corona Light	0.302	0.151	0.018	0.153	0.279	-5.795	0.183	0.048	0.037	0.113	0.039	0.065	0.171
(7) Heineken	0.269	0.145	0.018	0.131	0.311	0.108	-5.147	0.047	0.035	0.101	0.034	0.059	0.153
(8) Heineken Light	0.240	0.112	0.014	0.124	0.210	0.086	0.138	-5.900	0.026	0.089	0.028	0.051	0.135
(9) Michelob	0.301	0.140	0.015	0.146	0.208	0.089	0.135	0.042	-4.970	0.116	0.036	0.061	0.175
(10) Michelob Light	0.345	0.159	0.019	0.181	0.235	0.101	0.146	0.047	0.041	-5.071	0.046	0.072	0.196
(11) Miller Genuine Draft	0.346	0.159	0.019	0.182	0.235	0.101	0.146	0.047	0.040	0.130	-4.696	0.072	0.196
(12) Miller High Life	0.338	0.159	0.019	0.177	0.242	0.102	0.153	0.047	0.040	0.127	0.045	-3.495	0.191
(13) Miller Lite	0.344	0.159	0.019	0.180	0.237	0.101	0.148	0.047	0.040	0.129	0.046	0.071	-4.517
(14) Outside Good	0.016	0.007	0.001	0.009	0.011	0.005	0.006	0.002	0.002	0.006	0.002	0.003	0.009
<i>Cross-Elasticities by Category</i>													
6 Packs	0.307	0.152	0.018	0.155	0.275	0.104	0.180	0.047	0.038	0.115	0.039	0.065	0.174
12 Packs	0.320	0.154	0.019	0.163	0.250	0.102	0.161	0.047	0.039	0.121	0.042	0.068	0.183
24 Packs	0.356	0.160	0.019	0.189	0.222	0.099	0.136	0.047	0.041	0.134	0.048	0.073	0.201
Domestic	0.349	0.160	0.019	0.184	0.229	0.100	0.142	0.047	0.040	0.131	0.047	0.072	0.197
Imported	0.279	0.147	0.018	0.138	0.301	0.108	0.200	0.047	0.035	0.104	0.035	0.061	0.158

^aThis table provides the mean elasticities of demand for 12 packs based on the RCNL-1 specification (column (i) of Table IV). The cell in row i and column j is the percentage change in the quantity of product i with respect to the price of product j . Means are calculated across year-month-region combinations. The category cross-elasticities are the percentage change in the combined share of products in the category due to a 1 percent change in the price of the product in question. Letting the category be defined by the set $\mathcal{A}$, we calculate $\frac{\sum_{i \in \mathcal{A}} \epsilon_{ij}}{\sum_{i \in \mathcal{A}} \epsilon_{ij} + \epsilon_{jj}}$. The categories exclude the product in question. Thus, for example, the table shows that a 1 percent change in the price of a Bud Light 12 pack increases the sales of other 12 packs by 0.20 percent.

Read:

- The preferred RCNL demand models imply median own-price elasticities between about -4.33 and -6.10 .
- The market-price elasticities are much smaller, implying that most substitution is among beers rather than toward the outside good.
- Outside diversion is around **12%–17%** in the RCNL models.
- Table V shows that substitution is economically sensible: Bud Light buyers substitute heavily toward Budweiser, Miller Lite, Coors Light, and nearby package sizes, while Corona buyers substitute disproportionately toward Heineken and smaller packs.

5.5 Table VI and Table VII: supply estimates and markups

TABLE VI
BASELINE SUPPLY ESTIMATES^a

Demand Model: Data Frequency: Variable	Parameter	NL-1 Monthly (i)	RCNL-1 Monthly (ii)	RCNL-2 Quarterly (iii)	RCNL-3 Monthly (iv)	RCNL-4 Quarterly (v)
Post-Merger Internalization of Coalition Pricing Externalities	κ	0.374 (0.034)	0.264 (0.073)	0.249 (0.087)	0.286 (0.042)	0.342 (0.054)
Marginal Cost Parameters						
MillerCoors $\times$ PostMerger	γ_1	-0.608 (0.039)	-0.654 (0.050)	-0.649 (0.060)	-0.722 (0.042)	-0.526 (0.040)
Distance	γ_2	0.142 (0.046)	0.168 (0.059)	0.163 (0.059)	0.169 (0.060)	0.148 (0.049)

^aThis table shows the baseline supply results. We use the method of moments for estimation. There are 94,656 observations at the brand-size-region-month-year level in columns (i), (ii), and (iv) and 31,784 observations at the brand-size-region-year-quarter level in columns (iii) and (v). The samples exclude the months/quarters between June 2008 and May 2009. All regressions include product (brand $\times$ size), period (month or quarter), and region fixed effects. Standard errors clustered by region and shown in parentheses.

TABLE VII
BREWER MARKUPS FROM RCNL-1^a

Brand	6 Packs		12 Packs		24 Packs	
	Pre	Post	Pre	Post	Pre	Post
Bud Light	3.63	4.34	3.52	4.24	3.43	4.13
Budweiser	3.79	4.49	3.66	4.38	3.55	4.25
Coors	2.70	4.39	2.56	4.31	2.44	4.18
Coors Light	2.47	4.21	2.36	4.14	2.28	4.04
Corona Extra	3.30	3.18	3.04	2.91	3.04	3.03
Corona Light	3.02	2.91	2.75	2.65	2.87	2.80
Heineken	3.20	3.14	2.98	2.92	3.22	3.33
Heineken Light	2.87	2.81	2.61	2.50	2.75	2.69
Michelob	3.69	4.47	3.62	4.38	3.34	4.28
Michelob Light	3.61	4.34	3.53	4.23	3.46	4.06
Miller Gen. Draft	2.89	4.26	2.77	4.16	2.68	4.09
Miller High Life	2.91	4.28	2.80	4.20	2.74	4.13
Miller Lite	2.89	4.25	2.78	4.18	2.69	4.07

^aThis table provides the average markups for each brand-size combination separately for the pre-merger and post-merger periods, based on the RCNL-1 specification shown in column (ii) of Tables IV and VI.

Read:

- The central supply estimate is the post-merger conduct parameter κ . It is positive and statistically significant in every specification, ranging from about **0.249** to **0.342**. In the preferred RCNL-1 specification it is **0.264**.
- Interpreted literally, ABI and MillerCoors internalize about **one quarter to one third** of the pricing externality they impose on one another after the merger.
- Average brewer markups are about **\$3.60** per equivalent-unit product, or roughly **34%** of the retail price.
- Post-merger markups rise for domestic brands, especially for Miller and Coors, while import markups move little.
- The distance coefficient implies that shipping-related marginal costs account for roughly **2%–3%** of the retail price on average.
- The common MillerCoors cost shifter implies cost savings of about **\$0.66–\$0.70**, above and beyond pure distance reductions.

5.6 Figure 2 and Table VIII: efficiencies are real, but Nash–Bertrand needs implausible ABI cost increases

Read:

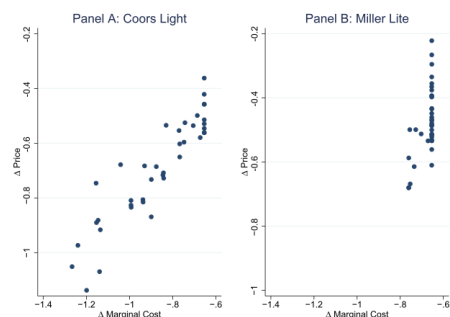


FIGURE 2.—Change in price against change in marginal cost. *Notes:* This figure plots the average regional difference in counterfactual price with no efficiencies and the observed price against the average regional difference in marginal cost for 12 packs of Coors Light and Miller Lite. Each dot is a region average in 2011 and is based on the RCNL-1 specification.

TABLE VIII
CHANGES IN ABI LOG COSTS WITH BERTRAND AND COORDINATION^a

	Budweiser	Bud Light	Michelob Light	Michelob Ultra
1 {Post-Merger and Bertrand}	0.122 (0.006)	0.120 (0.006)	0.089 (0.004)	0.102 (0.007)
1 {Post-Merger}	0.016 (0.014)	-0.002 (0.011)	0.124 (0.016)	0.050 (0.013)

^aThe dependent variable is log marginal costs from (i) the baseline model and (ii) an alternative with Nash-Bertrand pricing in all periods. The RCNL-1 specification is used to obtain the implied marginal costs. Observations are thus at the brand-size-region-month-scenario level. The estimation sample excludes observations from June 2008 through May 2009. All regressions include product (brand $\times$ size) fixed effects interacted with region fixed effects. Standard errors are clustered at the region level and shown in parentheses.

- Figure 2 shows that merger-related cost reductions pass through strongly into lower prices, with about **\$0.80** of price reduction per **\$1.00** of cost reduction for Coors Light.
- So the paper is not “anti-efficiency” — it finds substantial cost savings.
- Table VIII then asks what ABI costs would have had to do if post-merger competition were still Nash–Bertrand. For Budweiser, the answer is about **13.8%** higher costs, whereas the baseline model implies only about **1.6%**.
- That cost increase is viewed as implausible, which is one of the strongest arguments against post-merger Nash pricing.

5.7 Figure 3 and Table IX: the conduct shift is persistent and robust

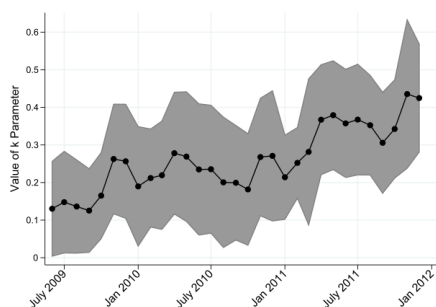


FIGURE 3.—Time-varying estimates of the κ parameter. *Notes:* This figure plots the point estimates of κ for each post-merger month-year combination, along with a 95% confidence interval. Estimation is carried out with the method of moments and we use the results generated with the RCNL-1 demand specification. There are 94,656 observations at the brand-size-region-month-year level. The sample excludes the months between June 2008 and May 2009. The marginal cost function includes the baseline cost shifters and product (i.e., brand $\times$ size), period, and region fixed effects. The instruments include the interaction of an indicator for ABI products with indicators for each post-merger time period. The confidence interval is calculated with standard errors clustered by region.

TABLE IX
SUPPLY-SIDE ESTIMATES WITH ALTERNATIVE PRE-MERGER NORMALIZATIONS^a

	(i)	(ii)	(iii)	(iv)	(v)
Post-Merger Internalization of Coalition Pricing Externalities	0.320 (0.066)	0.382 (0.058)	0.448 (0.051)	0.518 (0.043)	0.593 (0.036)
Pre-Merger Internalization	0.10	0.20	0.30	0.40	0.50

^aThis table shows the baseline supply results based on method-of-moments estimation. The results are generated with the RCNL-1 demand specification. There are 94,656 observations at the brand-size-region-month-year level. The sample excludes the months between June 2008 and May 2009. All regressions include the baseline marginal cost shifters, as well as product and time fixed effects. Standard errors are clustered by region and shown in parentheses.

Read:

- Figure 3 lets the conduct parameter vary over post-merger time. The point estimates generally **increase over time**, which is not what one would expect if the result were purely a temporary recession effect.
- Table IX changes the pre-merger conduct normalization. In every case, the post-merger conduct parameter rises significantly above the pre-merger normalization.
- So the paper’s core claim is robust: **something changed in post-merger conduct**.

5.8 Figure 4, Figure 5, and Table X: counterfactual merger decomposition

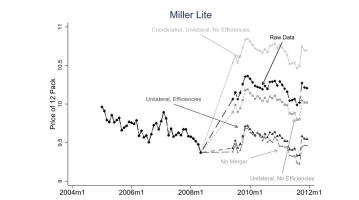


FIGURE 4.—Counterfactuals prices for Miller Lite. Note: This figure plots the average retail prices of Miller Lite 12 packs in the raw data and under four different counterfactual scenarios. Each dot represents the average prices across the 39 regions.

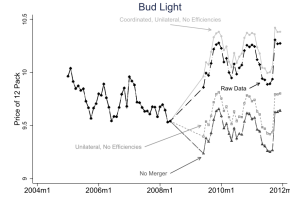


FIGURE 5.—Counterfactuals prices for Bud Light. Note: This figure plots the average retail prices of Bud Light 12 packs in the raw data and under three different counterfactual scenarios. Each dot represents the average prices across the 39 regions.

TABLE X RESULTS FROM COUNTERFACTUAL ANALYSIS ^a					
Coordinated Effects:	yes	yes	no	no	no
Unilateral Effects:	yes	yes	yes	yes	no
Efficiencies:	yes	no	yes	no	no
	(i)	(ii)	(iii)	(iv)	(v)
Retail Prices					
ABI	10.63	10.14	9.38	9.55	9.43
Miller	8.94	9.37	8.28	8.72	8.19
Coors	10.18	10.85	9.56	10.22	9.26
Brewer Markups					
ABI	4.45	4.56	3.81	3.97	3.84
Miller	4.52	4.32	3.83	3.63	3.05
Coors	4.25	4.06	3.61	3.41	2.08
Welfare Statistics					
Producer Surplus	22.1%	19.1%	10.3%	8.2%	—
ABI	10.3%	19.8%	-1.00%	9.3%	—
Miller	37.8%	20.2%	24.6%	9.1%	—
Coors	47.8%	12.9%	34.7%	3.5%	—
Consumer Surplus	-3.7%	-5.3%	-0.2%	-2.1%	—
Total Surplus	1.3%	-0.6%	1.8%	-0.1%	—

^aThis table provides volume-weighted mean prices and markups, separately, for 12-pack flagship brands of ABI, Miller, and Coors, under five different economic scenarios. Also shown are the percentage change in producer surplus, consumer surplus, and total surplus relative to the “No Merger” scenario. The welfare statistics are calculated using the complete data set (i.e., all products). Column (i) is based on the raw data and supply-side parameter estimates. Columns (ii) to (v) show the results from counterfactual scenarios. The numbers in column (ii) are computed assuming the merger occurs with coordinated and unilateral effects but without efficiencies. The numbers in column (iii) are computed assuming the merger occurs with unilateral effects and efficiencies but no coordinated effects. The numbers in column (iv) are computed assuming the merger occurs with unilateral effects but without efficiencies or coordinated effects. Lastly, the numbers in column (v) are computed assuming that the Miller/Coors merger does not occur. All statistics are for 2011.

Read:

- Figure 4 shows that for Miller Lite, the **unilateral effect and the efficiency effect are both large and nearly offset**. That is why the “No Merger” and “Unilateral + Efficiencies” paths look similar.
- Figure 5 shows that for Bud Light — an ABI brand — the difference between observed prices and the no-merger benchmark is almost entirely due to **coordinated effects**.
- Table X is the punchline. Relative to the no-merger benchmark:
 - observed ABI, Miller, and Coors prices are markedly higher,
 - producer surplus rises by **22.1%**,
 - consumer surplus falls by **3.7%**,
 - total surplus rises by **1.3%** because efficiencies are large.
- Without coordinated effects but with efficiencies, consumer surplus falls by only **0.2%**. So most of the consumer harm comes from coordination, not from unilateral effects alone.

6 Short summary

- Method.** Estimate a flexible beer demand system, plug it into a differentiated-products pricing model, and test whether a conduct parameter linking ABI and MillerCoors must be positive after the merger.
- Identification.** Use demand estimates to pin down unilateral incentives and use imports (Modelo and Heineken) as the outside comparison group to identify whether ABI prices move too much to be consistent with Nash–Bertrand.
- Application.** U.S. brewing industry, MillerCoors joint venture, 39 regions, 2001–2011 scanner data.
- Main result.** The data reject post-merger Nash–Bertrand competition. The merger appears to have facilitated coordination between ABI and MillerCoors, raising prices by about **6%–8%** above the Nash benchmark and markups by about **17%–18%**.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The MillerCoors joint venture cannot be understood purely as a standard differentiated-products merger with unilateral effects and efficiencies.
- To match the data, the model needs a post-merger increase in coordination between **MillerCoors and ABI**.
- The rival price increases are the crucial fact: they imply that the equilibrium concept itself changed after the merger.

7.2 Economic intuition

- The merger does two standard things: it raises unilateral pricing incentives between Miller and Coors, and it creates cost savings.
- In this market, those two forces are both important but approximately cancel in many products.
- What remains is softer competition between the merged firm and its leading rival, ABI.
- That softer competition is what drives the large sustained price increases and most of the loss in consumer surplus.

7.3 Takeaway

This paper shows that **mergers can change conduct, not just ownership**. A merger can create efficiencies and standard unilateral effects, but still harm consumers substantially if it also makes coordination easier. For antitrust, that means merger evaluation should not stop at unilateral simulation alone; it must also ask whether the merger changes how firms compete.